

Project Documentation

Triplify

Project Documentation

Travel Management Desktop Application

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Vision Document

Vision

Triplify aims to become the go-to personal travel management desktop application for passionate travelers. The application allows users to plan future trips, document them, and build a rich personal travel portfolio, all in one place.

The core business value is user engagement and retention through a deeply personal, data-rich travel management. By letting users save Places they love, connect them into Routes, organize everything into Trips, and capture memories as Stories with emotions and images, Triplify creates a travel journal that users return to repeatedly. The gamification layer (Badges and Statistics) further drives engagement and encourages users to explore more.

For the platform, this translates into a growing, active user base that can be monetized through a premium licence offering advanced features such as story management, calendar view for trips, and unlimited routes per trip.

Usage Scenario

Triplify targets individual travelers, from weekend adventurers to seasoned travelers, who want a structured, visual way to plan and remember their journeys. Today, these users scatter their travel data across saved locations in various apps, note-taking tools, spreadsheets, and photo galleries with no single source of truth.

A typical user discovers a place they want to visit, saves it, and gradually builds a collection of places across multiple countries. When a trip takes shape, they create a Trip, attach their saved Places, define Routes between them, and set dates. During or after the trip, premium users capture Stories tied to specific places or routes, tagging them with emotions and images. Over time, their profile accumulates statistics and unlocks achievement badges.

The application runs entirely on the user's local machine. All data, including places, trips, routes, stories, and images, is stored locally. A Configuration Manager role exists within the application to maintain shared reference data such as countries, categories, emotions, and badges, ensuring the platform stays curated and relevant.

Know your users

★ Standard User (Traveler)

Jano Novak is a travel enthusiast who takes 3 to 5 trips per year and is always dreaming about the next destination. Jano saves interesting places as they come across them, plans upcoming trips by organizing places into routes, and after each journey reviews his documented content. He checks the application several times a week for short sessions (5 to 15 minutes) to add a place or browse saved content, with heavier usage (30 to 60 minutes) during trip planning phases. Jano is not highly technical but is comfortable with modern desktop applications. Standard users can manage Places, Routes, Trips (with one Route per Trip), Tags, and view Badges and Statistics. They cannot create Stories or access the calendar view. There are expected to be thousands of standard users, forming the vast majority of the user base.

★ Premium User

Katarina Horvathova is a power traveler who documents every journey in detail. She uses all features intensively and values the ability to write Stories with emotions, descriptions, and image galleries for her Trip Places, Trip Routes, and Trips. She also relies on the calendar view to plan and manage her upcoming trips visually. Unlike standard users, Katarina can attach an unlimited number of Routes to a single Trip, which is essential for her complex multi-destination journeys. She spends 1 to 2 hours per week in the application and expects a polished, fast experience. Premium users are a smaller but highly engaged segment willing to pay for these additional capabilities.

★ Configuration Manager (Admin)

Martin Kovac is responsible for maintaining the application's reference data. He manages countries (including banning and unbanning), trip categories, emotions, badge groups, and badges. Martin uses the admin interface 2 to 3 times per week for short sessions (10 to 20 minutes), typically when new content needs to be added or moderated. He has deep knowledge of the application's data model. There are only 1 to 3 Configuration Managers.

Users language

★ Place

A real-world location that a user saves. It has a title, description, geographic coordinates, a country, and an optional cover image. A user may have dozens to hundreds of places. Places are the fundamental building blocks of the entire system. They feed into Routes, Trips, and the Map.

★ **Route**

An ordered sequence of at least 2 Places that forms a travel path. A route has a calculated length based on the coordinates of its places. Users typically have 5 to 30 routes. The same place cannot appear consecutively in a route.

★ **Trip**

A travel plan or documented journey. A trip belongs to one or more countries, has a category, optional tags, a date range, and a status (planned, ongoing, visited, canceled). Trips are the main organizational entity. Standard users can attach one Route per Trip, while premium users can attach unlimited Routes. An active user may have 10 to 50 trips.

★ **Trip Place**

A Place within the context of a specific Trip. It carries trip-specific data: visit date, status, and images from that particular journey. This allows the same Place to appear in multiple trips with different associated data each time.

★ **Trip Route**

A Route within the context of a specific Trip, carrying trip-specific status and images. The user can rearrange places within a trip route, with the option to update the original route or create a new one.

★ **Story (Premium only)**

A personal narrative or memory from a trip, available exclusively to premium users. Stories can be linked to a Trip, Trip Place, or Trip Route. Each story can have an emotion, description, tags, and multiple images with descriptions. Stories are the journaling heart of the application.

★ **Country**

A predefined geographic entity with a name and flag emoji. Countries can be banned by admins, making them unavailable for new trips and places. Users interact with countries as filters and classifiers.

★ **Category**

A classification for Trips (e.g. Adventure, Cultural, Beach, Road Trip). Categories are managed by the Configuration Manager and have a name, description, icon, and color. Each trip has exactly one category.

★ Tag

A user-created label for organizing Trips and Stories. Tags are unique per user, have a name and a randomly assigned color. Tags enable cross-cutting filtering across trips and stories.

★ Emotion

A predefined feeling indicator (e.g. Excited, Relaxed, Amazed) managed by the Configuration Manager. Premium users attach an emotion to a Story to express how they felt at that moment. Represented by a name and emoji icon.

★ Badge / Badge Group

Badges are achievements users earn by reaching milestones (e.g. "Visited 10 countries"). Badges belong to Badge Groups, are ordered by level, and each has a required value threshold. Users see locked/unlocked states and progress toward the next badge.

★ Image

A photo attached to a Trip, Trip Place, Trip Route, or Story, with an optional description. Images can also serve as cover photos for Places, Routes, and Trips. Loaded with pagination, ordered by creation time.

★ Statistic

An automatically updated counter reflecting the user's activity: countries visited, total trips, places visited, travel days, photos uploaded, and stories created (premium only).

Most common tasks

As a standard user, **Jano Novak** wants to:

- ★ Quickly add a new Place with a location, country, and optional photo, so that interesting spots are captured the moment they are discovered.
- ★ Browse saved Places on an interactive Map with clustering, so that the spatial distribution of saved locations is immediately visible.
- ★ Create a Trip by choosing countries, a category, tags, and dates, so that an upcoming journey is structured from the start.

- ★ Add existing Places to a Trip (as Trip Places) and set their visit status and images, so that the trip is fleshed out with concrete stops.
- ★ Define a Route by connecting multiple Places in order and see the calculated length, so that a logical travel path is planned.
- ★ Add one Route to a Trip and view it in the trip context, so that the planned path is documented alongside the trip details.
- ★ Update the Status of a Trip or Trip Place (planned, ongoing, visited), so that progress is tracked in real time.
- ★ Filter Trips by country, category, tags, status, or date, so that a specific trip can be found quickly in a growing list.
- ★ View personal Statistics (countries visited, total trips, travel days, etc.), so that overall travel achievements are visible at a glance.
- ★ View collected Badges and progress toward locked ones, so that there is motivation to keep exploring.
- ★ Search Places and Trips by title or description, so that specific content is found without scrolling.

As a premium user, **Katarina Horvathova** wants to:

- ★ Create, update, and delete Stories linked to a Trip, Trip Place, or Trip Route, with an emotion, description, and multiple images, so that personal travel memories are preserved with rich context.
- ★ View all Stories related to a specific place, route, or trip, so that the full narrative of a journey is easy to revisit.
- ★ View planned trips in a Calendar view, so that upcoming journeys are easy to visualize and manage in terms of scheduling.
- ★ Add an unlimited number of Routes to a single Trip, so that complex multi-destination journeys are fully documented.
- ★ Access the "Stories Created" statistic, so that content creation output is tracked alongside other travel metrics.

As a configuration manager, **Martin Kovac** wants to:

- ★ View, add, update, and delete Trip Categories with name, description, icon, and color, so that the classification system stays relevant.
- ★ View all Countries, ban or unban them, and add new countries, so that the available country list matches platform policies.
- ★ Update and delete only countries that he personally created, so that shared predefined data is protected.
- ★ View, add, update, and delete Emotions, so that users have a set of feelings to choose from.
- ★ View, add, update, and delete Badges and delete Badge Groups, so that the gamification layer evolves over time.

Main Processes

The primary business process in Triplify follows the natural lifecycle of a journey, from discovery through planning, traveling, and finally documenting. The process spans both the User and the Configuration Manager roles.

Process 1: Trip Lifecycle (User)

1. Discover and Save Places - The user finds interesting locations and saves them with a country, coordinates, and optional details.
2. Build Routes - The user connects saved places into ordered routes with calculated distances.
3. Plan a Trip - The user creates a trip, selects countries, a category, tags, and date range.
Status: Planned.
4. Attach Places and Routes to the Trip - The user adds Trip Places and Trip Routes.
5. Travel - The user updates statuses to Ongoing and logs visit dates as the trip progresses.
6. Document - the user adds Images with memories for the individual Trip or Trip Place. The premium user writes Stories linked to Trip Places or Trip Routes, attaching emotions, tags, and photos.
7. Complete - The user marks the trip as Visited. Statistics and badge progress update automatically.

Exception Flow - Trip Cancellation: The user decides not to go on the trip. He update the trip status to "canceled". This skips the "Travel" and "Document" steps and means the trip does not count toward standard completion statistics.

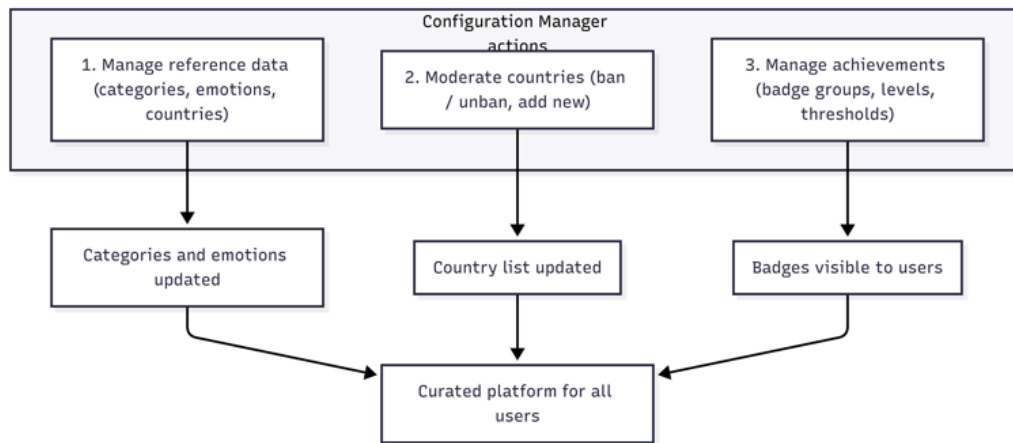
Alternative Flow: While traveling (Status: Ongoing), the user changes their mind. The user can rearrange places within a trip route or add completely new places.



Process 2: Platform Curation (Configuration Manager)

1. Manage Reference Data - The CM adds and updates categories, emotions, and countries as the application evolves.
2. Moderate Countries - The CM bans countries that should not be available, choosing a status for existing records.
3. Manage Achievements - The CM creates badges to a specific badge group with level thresholds to drive user engagement.

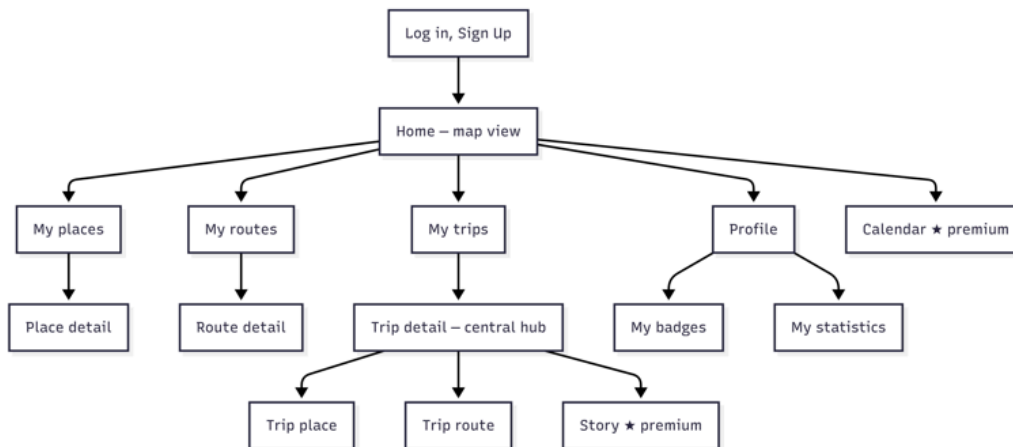
Exception Flow: The CM want to delete a country that was standard or created by different CM. The application shows error notification that it is not possible and suggests to ban the country.



Navigation

The application has two distinct navigation structures depending on the user role.

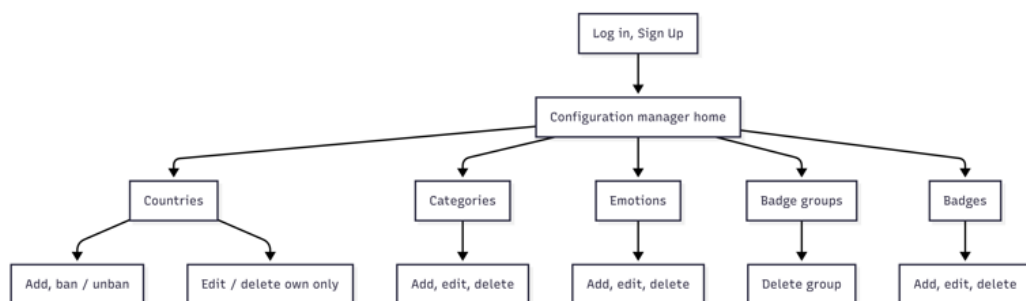
Standard User Navigation



Level	Page / Section	Actions
Home (Map)	Interactive map with all Places, Routes, Trips	Filter by object type, zoom/cluster
My Places	List of all user's places with filters	Add, Edit, Delete, View on Map
Place Detail	Single place with images and linked trips/routes	Edit, Delete
My Routes	List of all user's routes	Add, Edit, Delete
Route Detail	Route with ordered places and length	Rearrange Places, Edit, Delete
My Trips	List of trips with filters and ordering	Add, Edit, Delete, Change Status

Trip Detail	Trip overview with Trip Places, Trip Routes, Stories (premium)	Add/Remove Places and Routes, Write Story (premium)
Trip Place	Place in trip context with images	Edit, Upload Images, Change Status
Trip Route	Route in trip context with images	Rearrange, Upload Images, Change Status
Story (premium)	Story with emotion, tags, images	Edit, Delete, Add Images
Calendar (premium)	Calendar view of planned trips	View and navigate planned trips
Profile	User profile and avatar	Edit Profile, Change Password, Delete Account
My Badges	Badge groups with locked/unlocked badges	View progress
My Statistics	Dashboard of travel counters	View only (auto-updated)

Configuration Manager Navigation



Page	Content	Actions
Countries	List of all countries with ban status	Add, Ban/Unban, Edit/Delete own
Categories	List of trip categories	Add, Edit, Delete
Emotions	List of all emotions	Add, Edit, Delete

Badge Groups	List of badge groups with badges	Delete Group
Badges	Badges within groups by level	Add, Edit, Delete

IMPORTANT NOTE: The Trip is the central hub of the user experience. From a Trip page, the user accesses Trip Places, Trip Routes, and Stories. Navigation between these related entities should be seamless: creating a Story from a Trip Place, adding a Place to a Route from within the Trip, etc. The Map serves as the primary home view, providing a spatial entry point to all content.

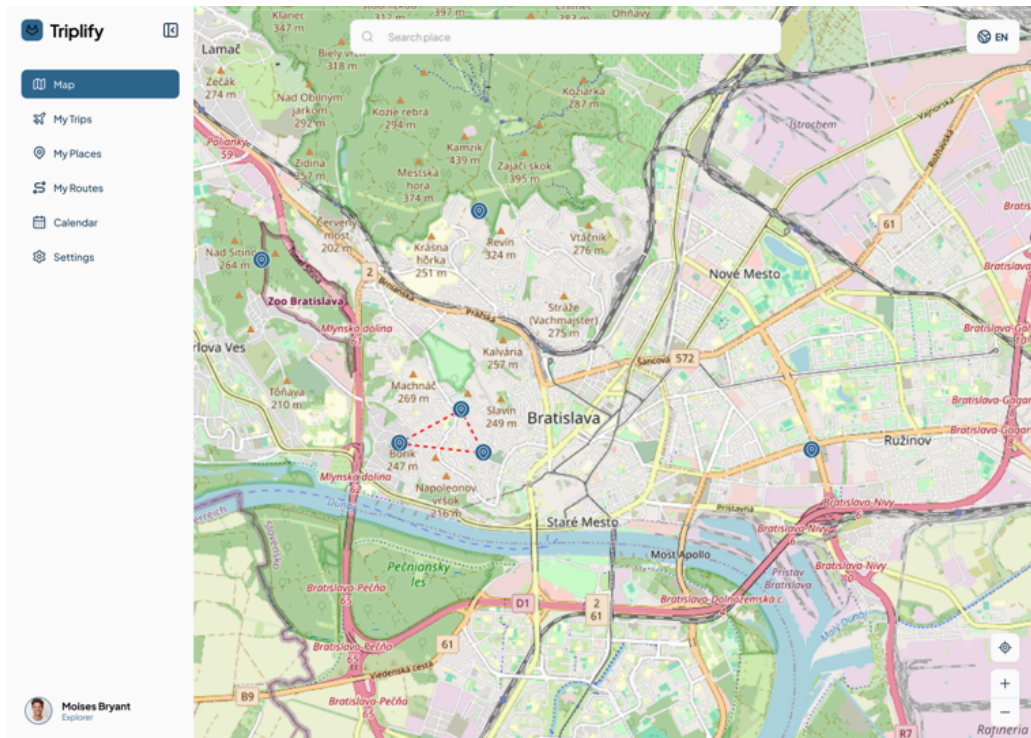
Mockups

Below are the key homepage elements and top screens for each role.

Standard User - Home Page (Map View)

The home page is an interactive map displaying the user's saved Places, Routes, and Trip markers. Key elements:

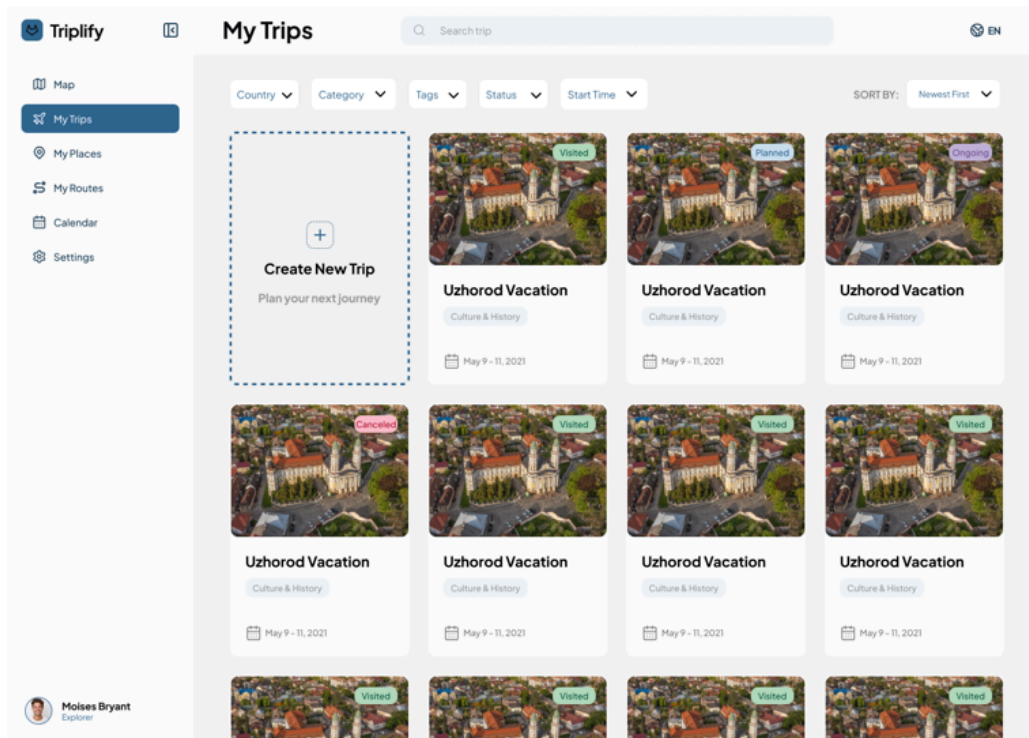
- ★ Interactive Map - Shows all user's Places as pins, with clustering when zoomed out. Routes displayed as connected lines. Clicking a marker opens a quick preview.
- ★ Left Navigation Menu - Links to Map, My Trips, My Places, My Routes, Calendar, Settings, and Profile.
- ★ Quick-add Button - Options to create a New Place or a New Trip.



Map page

Standard User - My Trips Page

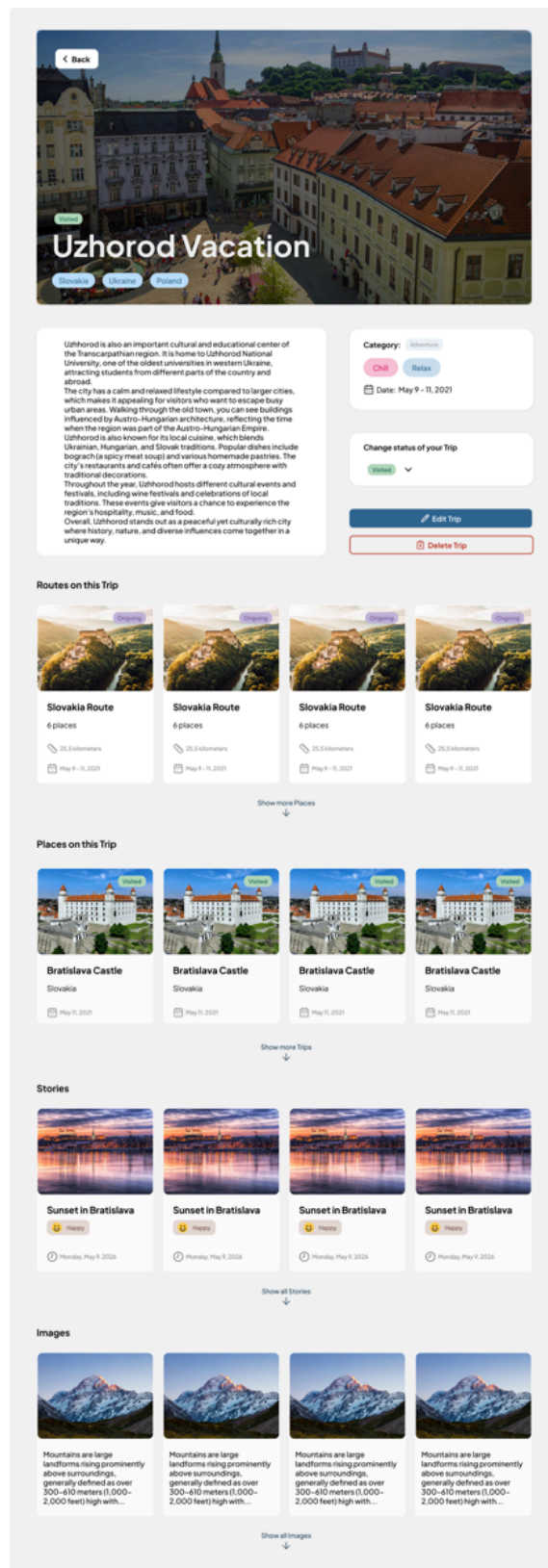
- ★ Trip Cards - Each card shows cover image, title, date range, status badge, category icon, and country.
- ★ Filter Panel - Filter by country, category, tags, status, and start time. Order by start time.
- ★ Pagination - Dynamic loading of trip cards as the user scrolls.
- ★ Create Trip Button - Add button to start a new trip.



My Trips Page

Standard User - Trip Detail Page

- ★ Trip Header - Cover image, title, dates, status selector, category, country flags, and tags.
- ★ Trip Places Section - Ordered list of Trip Places with status, visit date, and thumbnail. Drag-to-reorder supported.
- ★ Trip Routes Section - List of Trip Routes with length and place count. Standard users see one route, premium users see all.
- ★ Stories Section (Premium) - Timeline of Stories linked to this trip, showing emotion, time, and preview.
- ★ Image Gallery - All images from Trip Places and Trip Routes aggregated.

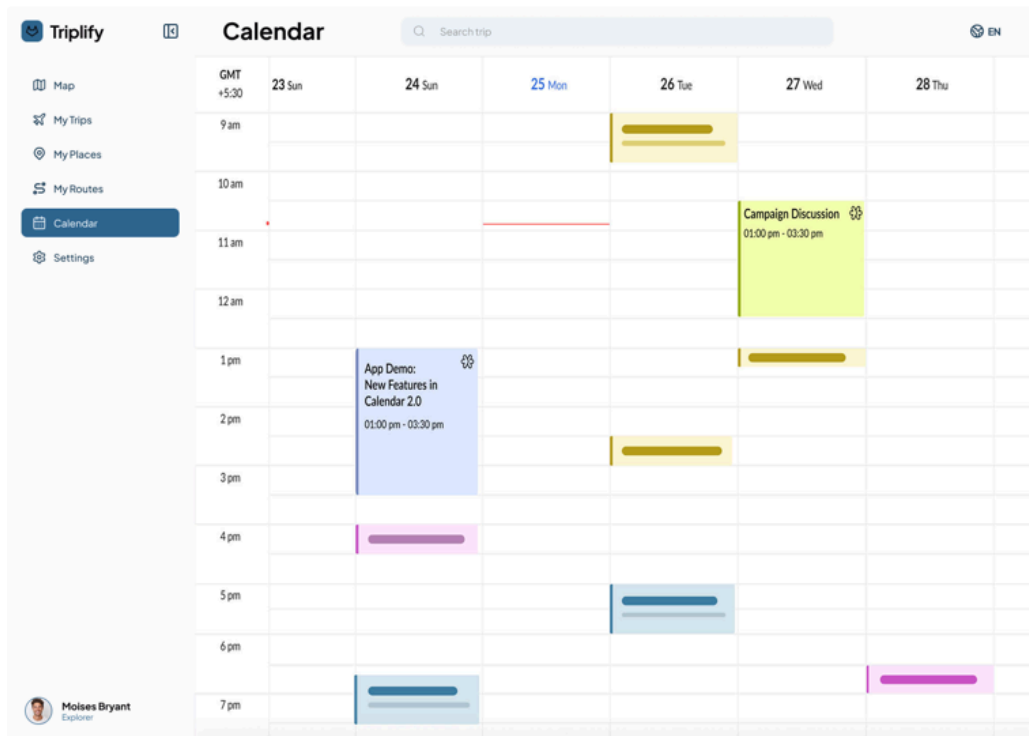


Trip Details Page

Premium User - Calendar View

★ Monthly/Weekly Calendar - Trips displayed as date-range blocks on a calendar grid. Color-coded by status.

- ★ Trip Preview on Hover - Hovering over a trip block shows title, countries, and status.
- ★ Navigation Controls - Switch between months and weeks. Quick jump to current date.



Calendar Page

Configuration Manager - Home Page (Dashboard)

- ★ Countries Overview - Total countries count, active vs. banned. Quick link to manage.
- ★ Categories Overview - List of current categories with usage count.
- ★ Emotions Overview - Grid of current emotions with icons.
- ★ Badges Overview - Badge groups count, total badges. Quick link to manage.
- ★ Left Navigation Menu - Links to Countries, Categories, Emotions, Badge Groups, Badges.

Non functional requirements

★ Desktop Application

Triplify is a desktop application. The user interface should be designed for comfortable use on standard desktop and laptop screens. The layout should make good use of the available screen space, particularly for the Map view and Calendar view. JavaFX will be used with JDK 25. The application should work on Windows, macOS and popular Linux distributions

★ Local Data Storage

All application data is stored locally on the user's machine. There is no remote server or cloud component. The application must handle local file storage efficiently, including images, user profiles, and all travel-related data. Data integrity and reliable persistence are essential. For local storage, SQLite should be used. The application should allow migration of data between the versions using custom migration service.

★ Image Handling

The system must support image uploads up to 10 MB per file. Cover images and gallery views must load progressively.

★ Security and Authentication

Since all data is stored locally, the application must protect user data on the local machine. Authentication via username and password is required to access the application. Passwords must meet basic strength requirements (Bcrypt should be used as a hash function). Role-based access control separates standard users, premium users, and configuration managers within the local environment.

★ Localization

The application must support Slovak and English languages. All user-facing labels, statistic titles, system messages, and interface elements must be available in both languages. The user should be able to switch between languages in the application settings. ResourceBundle will be used.

★ Data Privacy

Users must agree to a Privacy Policy and Terms of Service during registration. The application must allow users to delete their accounts and all associated data from the local storage.

★ Usability

The interface should follow standard desktop UX conventions and provide an intuitive, easy-to-learn experience. Common actions like creating places, managing trips, and navigating between related entities should require minimal effort. The layout and interaction patterns should be consistent across all pages of the application.

★ Performance Requirements

- When starting the application on the first time, the loading should be done in less than 20 seconds. When starting application for the second and next times, the loading should be done

in less than 5 seconds on baseline hardware (standard dual-core processor, 8GB RAM). The navigation between pages should be executed up to 2 seconds.

- When viewing the "Home Page (Map View)" that has up to 5,000 saved Places, the application must maintain a minimum UI frame rate of 30 FPS during moving and zooming on baseline hardware (standard dual-core processor, 8GB RAM).

★ Logging

For the excellent debugging experience logging should be implemented. With the logging of individual actions and exceptions. SLF4J will be used with standard logging levels (DEBUG, INFO, WARN, ERROR).

★ Validation and Exceptions handling

The application must clearly show validation errors to the user in appropriate way. Additionally, not allowed actions should be also prohibited and displayed appropriately for the user. The application should be secured from SQL-injections and other attacks.

Business model

Costs

The financial expenses are divided into two sections: Initial development cost and Annual support:

- **Initial development cost:** 30,000€ (one-time investment to build the core platform, MVP, and Premium features).
- **Annual support and maintenance:** 7,000€ / year - covers system updates and support.
- **Total Year 1 Cost:** 37,000€

Revenue

The application utilizes a Freemium to Premium conversion model. The core application is free to drive mass user, while for advanced planning and documentation features the subscription (licence) is needed. Monetization strategy is Premium subscription with a price of 5€/month for the licence. It is expected, that 1000 licences in the first year are enough to cover all the expenses.

RACI matrix

Role	Project Manager	Software Architect	Backend Developer	Backend Developer	Frontend Developer	Frontend Developer	UI/UX Designer	QA Tester	QA Tester
Person/Project part	Heorhii Sladkovskiyi	Yurii Oliinyk	Maksym Ananchenko	Brekhov Dmytro	Volodymyr Sorochynskiyi	Malokhatko Andrii	Arseniy Malyuk	Diurd Vitalii	Dmytro Shapovalov
Application architecture management	A	R	I	C	I	I			
Art direction	A				I	I	R	C	
Interaction design	A			I	I	I	R	C	
Database structure management	C	A	R	R					
Server environment management	A	R	C	I					
Data persistence	C	A	R	R					
Data migration	I	A	R	I					
Business-logic implementation	A	I	R	R					
Third-Party integration	A	R	I	C					
UI Implementation	A				R	R	C		
UI development quality management	A				R	I	C		
Security analysis	A					C		R	I
System testing	A	C						I	R
Penetration testing	A	C				C		R	I
Performance testing	A					C		I	R

- R - Responsible
- A - Accountable
- C - Consulted
- I - Informed

Documentation

All of the information from the Vision Document were successfully implemented. The application uses JavaFX framework for the ui part. The gradle project is divided into 5 subprojects:

- domain - core project of the application. Has repository interfaces, entities and domain utils
- infrastructure - data-access layer of the application. Has implementations of repositories and storage services
- application - buiseness-logic of the application. Has services, business-logic and other utils
- ui - JavaFX project for the ui part of the application
- bootstrap - wires all of the projects, orchistrates loading, executing of dependency injection configurations and other global wire actions. The usage of bootstrap allows to decouple dependencies and improves quality of the solution

The application also has real-time localization for all of the possible actions. English and Slovak are supported. As an SQL database engine, SQLite is used with an Spatialite extension (needed for working with coordinates and map). HikariCP is used for improving performance of JDBC connections.

There are three types of saved data withing the application:

- saving data to DB (mainly)
- images (to %APPDATA%/Triplify/Images)
- session xml file (%APPDATA%/Triplify/session.xml)
- licences (%APPDATA%/Triplify/licences/) in json format, with signing

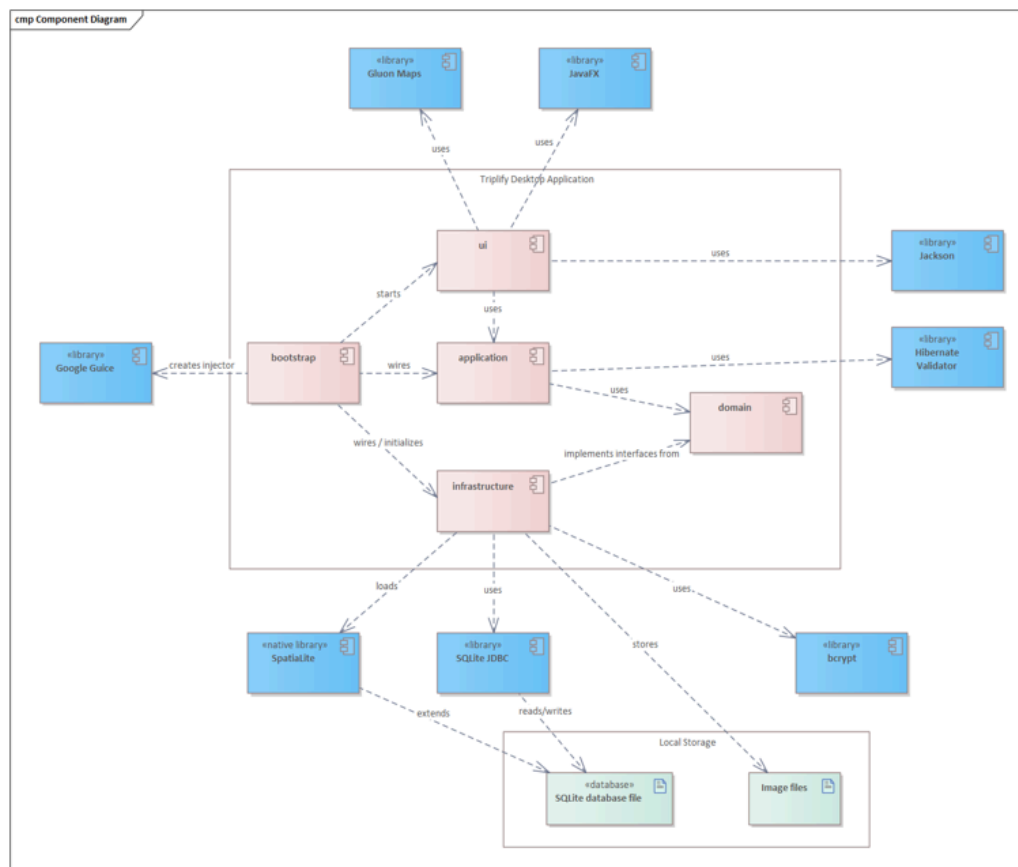
Links

Repository with the solution:  [GitHub - vava-2026/triplify](https://github.com/vava-2026/triplify)

Design in Figma:  [VAVA-Project](#)

Diagrams

Component Diagram

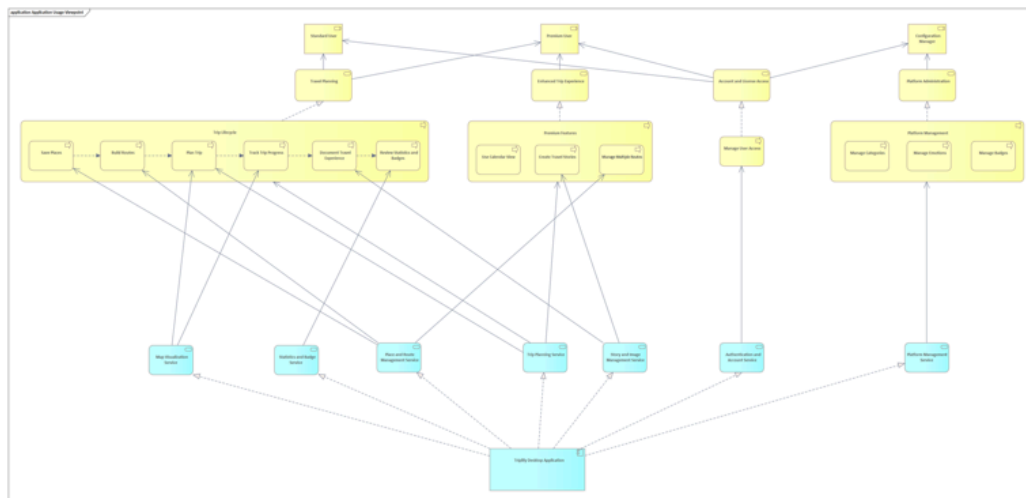


UML Component Diagram

The UML component diagram shows the internal structure of the Triplify desktop application. The application is divided into separate modules: bootstrap, UI, application, domain, and infrastructure. The bootstrap module starts the application and initializes dependency wiring. The UI module represents the user interface and communicates with the application layer. The application module contains the main use-case logic and coordinates operations between the UI, domain, and infrastructure parts. The domain module contains the core business concepts, domain logic, important abstractions, and repository/service interfaces. The infrastructure module implements technical details such as persistence, storage, and integration with local database functionality.

The diagram also separates the application from supporting external libraries and local storage. External libraries include JavaFX and Gluon Maps for the UI, Google Guice for dependency injection, Jackson for data processing, Hibernate Validator for validation, bcrypt for password hashing, SQLite JDBC for database access, and Spatialite for spatial database support. Local storage is represented by the SQLite database file and stored image files used by the application.

Application Usage Viewpoint

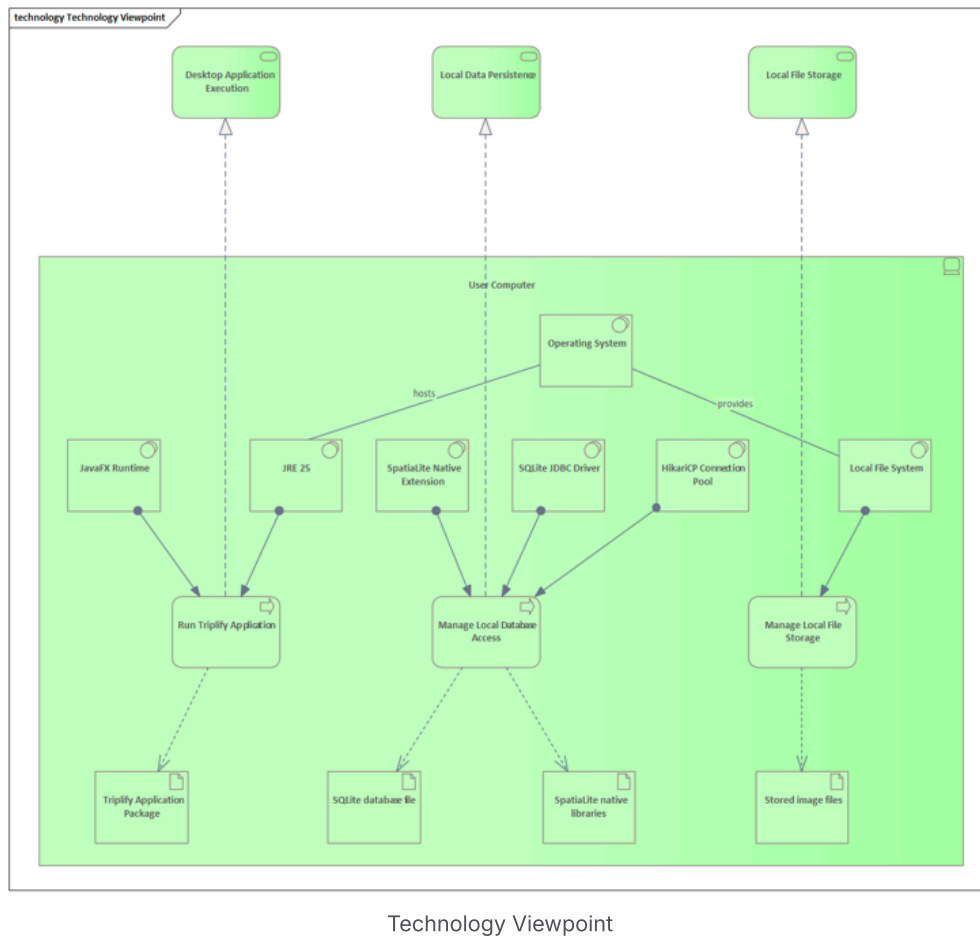


Application Usage Viewpoint

The Application Usage Viewpoint shows how the user roles use Triplify and which application services support their activities. There are 3 roles: Standard User, Premium User, and Configuration Manager.

Standard users are supported by the travel planning part of the application, which includes saving places, building routes, planning trips, tracking progress, documenting travel experiences, and reviewing statistics and badges. The Premium User extends the standard travel planning functionality with additional premium features, such as calendar view, travel stories, and multiple routes. The Configuration Manager uses the platform administration part of the application to manage platform data, including categories, emotions, and badges. The diagram also includes account and license access, which supports all user roles and is handled by the authentication and account service. Application services are shown as the bridge between business processes and the Triplify Desktop Application.

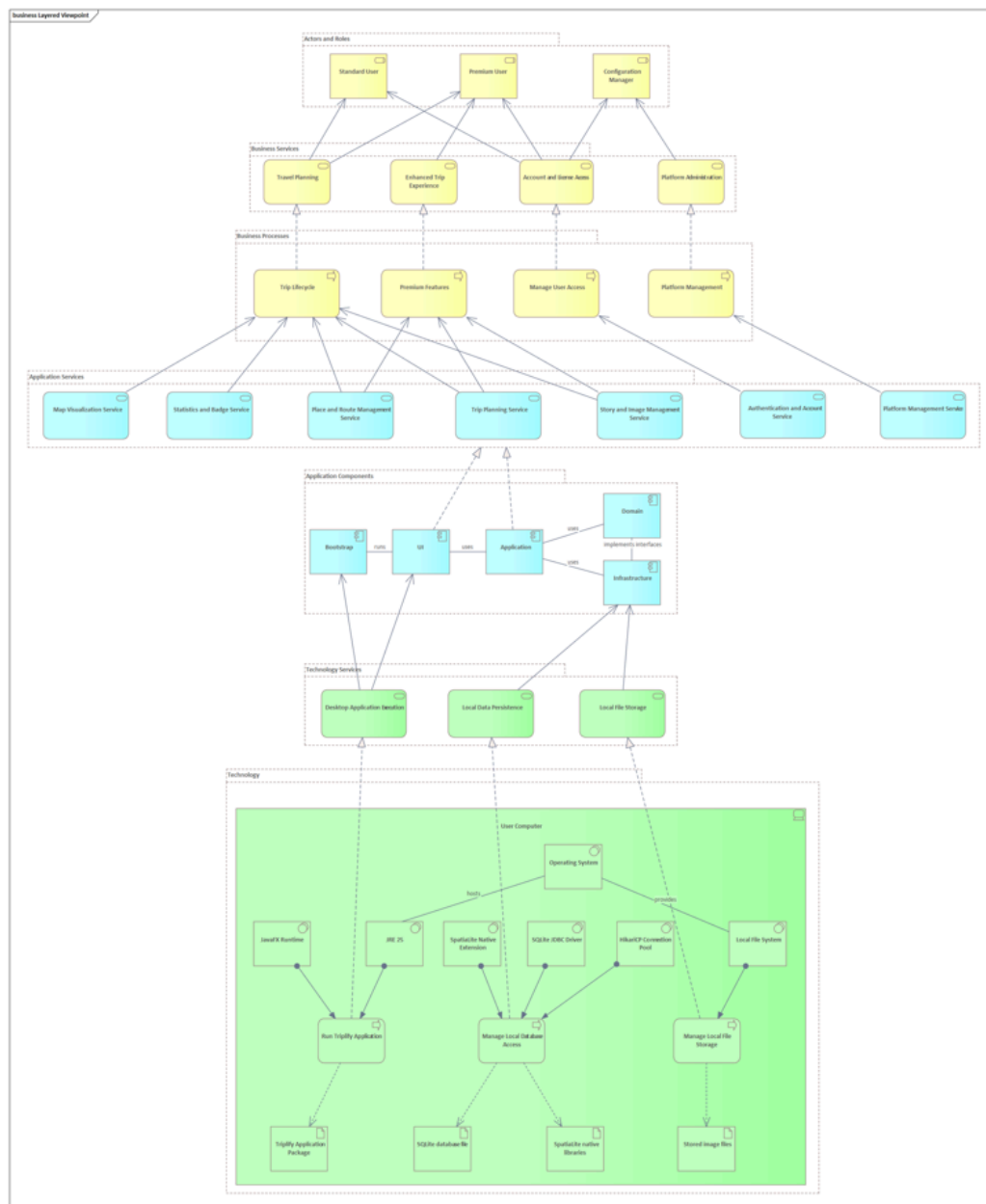
Technology Viewpoint



The Technology Viewpoint shows the technical environment in which the Triplify desktop application is executed. The application is placed on the user computer as a Triplify application package and runs as a Java desktop application. The diagram also shows the main runtime parts used by the application, including JRE 25 and JavaFX runtime.

The diagram separates the technical support into three main technology services: desktop application execution, local data persistence, and local file storage. Desktop application execution represents the process of running the Triplify application package. Local data persistence is handled through SQLite JDBC, HikariCP connection pool, and the SpatiaLite native extension - these components are related to access to the local SQLite database file and the SpatiaLite native libraries used for spatial functionality. Local file storage is shown separately from database persistence. It is provided by the local file system and is used for storing image files on the user's computer.

Layered Viewpoint



Layered Viewpoint

The Layered Viewpoint gives an overall view of the Triplify architecture from business roles down to the local technical environment. The top part shows the main roles in the system: Standard User, Premium User, and Configuration Manager. These roles use business services such as travel planning, enhanced trip experience, account and license access, and platform administration.

The business processes describe how these services are provided. Trip lifecycle covers the main travel planning flow, premium features extend this functionality for premium users, manage user access represents authentication and license-related access, and platform management covers administration of platform data. These processes are supported by application services

such as trip planning, place and route management, map visualization, story and image management, authentication, and platform management.

The application components layer shows the internal structure of the application using the main project modules: bootstrap, UI, application, domain, and infrastructure. The bootstrap module starts the application, the UI module communicates with the application layer, the application module coordinates use cases, the domain module contains core business logic and abstractions, and the infrastructure module provides technical implementations.

The lower part of the diagram connects the application architecture with the local technical environment. It shows that Triplify runs on the user computer and uses desktop execution, local database persistence, and local file storage as its main technology services. The technology layer includes the Java runtime and JavaFX for running the desktop application, SQLite JDBC, HikariCP, and SpatiaLite for database access, and the local file system for storing application files such as images.

Project Execution

Run the app using:

```
1 | ./gradlew :bootstrap:run
```

Configure git lfs: [Git LFS](#)

Seed user

The application has users with seeded data

Pro User

- **email:** moises@gmail.com
- **password:** 12345678

Builds

There are two types of builds.

1. /clean/* - clean jar, without seeded data for testing (without demo user)
2. /seeded/* - seeded version of application, contains demo user for fast testing.

Demo user

Pro User

- **email:** moises@gmail.com
- **password:** 12345678

Execution

If you want to have clean install or install older version of the application than current, remove everything inside `%APPDATA%/Triplify/*` folder.

Each of modules contains jar executable. There are strict prerequisites:

- **JDK 25** or higher is required

For executing the program, execute `run.bat` or `run.sh` as administrator, depending on your OS:

- Windows - `run.bat`
- Linux or mac - `run.sh`
- On mac with arm processor, run `brew install --cask zulufx` before execution

You can also directly execute `.jar` file with command:

Windows:

```
1 @echo off
2 java --enable-native-access=ALL-UNNAMED -jar "triplify.jar"
3 pause
```

Linux or mac:

```
1 #!/bin/sh
2 brew install --cask zulufx
3 java --enable-native-access=ALL-UNNAMED -jar "triplify.jar"
```